

JAEEYON JO, Ph.D.

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RESEARCH PROFILE

Electron microscopist specializing in monochromated STEM-EELS, electron tomography, and nanoscale mapping of structure, chemistry, and functional response. Experienced with monochromated low- and core-loss EELS, 4D-STEM, atomic-resolution TEM/STEM, and collaborative characterization of catalytic and electrocatalytic materials and interfaces. Current interests include extending advanced electron microscopy and spectroscopy toward reaction-relevant, cryogenic, magnetic, and quantum materials.

EDUCATION

Seoul National University, Seoul, Republic of Korea 2018-2025
Ph.D., Materials Science and Engineering

Dissertation: Nanoscale Near-Fields Mapping in a Chiral Plasmonic Nanoparticle via Low-loss EELS in a TEM (Advisor: Prof. Miyoung Kim)

Seoul National University, Seoul, Republic of Korea 2014-2018
B.S., Materials Science and Engineering, Summa Cum Laude (GPA 4.18/4.3)

RESEARCH EXPERIENCE

University of Illinois Chicago Chicago, IL
Postdoctoral Research Associate Sep. 2025-Present

- Contribute to commissioning and experimental implementation of a newly installed monochromated, magnetic-field-controlled TEM/STEM platform, working directly with vendor engineers and instrument developers to establish reliable imaging and spectroscopy capabilities.
- Exploring monochromated TEM/STEM, EELS, and 4D-STEM approaches to quantum and functional materials, with ongoing efforts to extend field-controlled microscopy and atomic-scale spectroscopy toward magnetic materials.

Seoul National University Seoul, Republic of Korea
Postdoctoral Researcher Feb. 2025-Aug. 2025

- Continued advanced electron spectroscopy and microscopy studies of plasmonic and functional nanomaterials, including monochromated EELS and collaborative materials characterization.
- Collaborate on polarization-controlled photon-induced near-field electron microscopy (PINEM) studies of chiral plasmonic nanostructures, connecting orientation-dependent optical excitation with spatially resolved electron microscopy.

Seoul National University Seoul, Republic of Korea
Graduate Researcher, Materials Science and Engineering Mar. 2018-Feb. 2025

- Established monochromated low-loss EELS tomography to directly map three-dimensional plasmonic near-fields in a chiral nanoparticle and connect nanoscale electromagnetic response with circular dichroism.
- Investigated optical and electronic responses of heterogeneous nanostructures using monochromated EELS, including plasmon-mode evolution in metal/dielectric systems.
- Contributed electron microscopy and spatially resolved EELS characterization to collaborative studies of catalytic and electrocatalytic materials, including catalyst reconstruction, electrode interfaces, local chemistry, and oxidation-state-sensitive analysis.

VISITING EXPERIENCE

Ernst Ruska-Centre, Forschungszentrum Juelich Juelich, Germany
Visiting Researcher Oct. 2020-Mar. 2021

- Participated in the Q-SORT project on orbital-angular-momentum-resolved EELS and performed STEM tomography comparing HAADF-based and secondary-electron-based three-dimensional reconstructions.

Electron Microscopy & Materials Science Laboratory, The University of Tokyo Tokyo, Japan

Visiting Researcher Aug.-Sep. 2025

- Advanced education for the magnetic-field-free atomic-resolution microscope instrument.

SELECTED EXPERIMENTAL & ANALYTICAL SKILLS

Electron microscopy & spectroscopy: Monochromated low-loss and core-loss EELS; STEM-EELS spectrum imaging; ELNES/oxidation-state-sensitive analysis; HAADF, ABF, and iDPC STEM; **atomic-resolution TEM/STEM; electron tomography;** 4D-STEM.

cryogenic holder-based TEM experiments; FIB lamella preparation; nanoparticle dispersion; mechanical polishing.

Instrumentation: Thermo Fisher Themis Z; JEOL Neo-ARM; JEOL MARS; aberration correction and advanced imaging/spectroscopy workflows.

Data analysis: MATLAB (proficient); Python (basic scripting/data analysis); EELS, tomography, and 4D-STEM analysis.

PUBLICATIONS

- P1.** "Helicity-Resolved Spatiotemporal Mapping of Chiral Plexcitons in Helicoids", Han, J. H.; Ramesh, S.; **Jo, J.**; Chabera, P.; Kim, R. M.; Cho, S. H.; Ha, I. H.; Sahu, A.; et al., arXiv preprint arXiv:2606.09097, 2026.
- P2.** "Comprehending Non-Monochromatic Plasmonic Behavior of Metal Nanoparticles on Dielectric Nanowires using Monochromated EELS", **Jo, J.**; Ali, A.; Han, J. H.; Kwon, J.-H.; Nam, K. T.; Yi, G.-C.; Kim, M., Physical Review Materials 2025, 9, 125202.
- P3.** "Bioinspired Lignin-Aluminosilicate Composite for CO₂ Storage", Koo, J. W.; Yeo, J. B.; Jang, J. H.; **Jo, J.**; Heo, J.; Yoon, J.; Jo, Y. I.; Ko, C.; Lee, M. Y.; Hong, J. S.; Kim, M.; Nam, K. T., ACS Omega 2025, 10(31), 35134-35145.
- P4.** "Unveiling the reconstruction of copper bimetallic catalysts during CO₂ electroreduction", Kim, I.; Lee, G. B.; Kim, S.; Jung, H. D.; Kim, J. Y.; Lee, T.; Choi, H.; **Jo, J.**; et al., Nature Catalysis 2025, 8(7), 697-713.
- P5.** "Isotropic Size Control of Chiral Gold Helicoids", Han, J. H.; Ha, I. H.; Cha, J. H.; **Jo, J.**; Lv, J.; Im, S. W.; et al.; Nam, K. T., The Journal of Physical Chemistry C 2025, 129(14), 7020-7030.
- P6.** "Direct three-dimensional observation of the plasmonic near-fields of a nanoparticle with circular dichroism", **Jo, J.**; Ryu, J.; Huh, J.-H.; Kim, H.; Seo, D. H.; Lee, J.; Kwon, M.; Lee, S.; Nam, K. T.; Kim, M., ACS Nano 2024.
- P7.** "Thin Titanium Nitride Layer-Inserted Porous Ru_{0.3}Sn_{0.35}Ti_{0.35}O_{2-x} Anode for Stable Chlorine Evolution Reaction", Choi, W. I.; **Jo, J.**; Choi, S.; Hong, J. S.; Yoo, I.; Randriamahazaka, H.; Kim, M.; Park, S.; Nam, K. T., ACS Applied Energy Materials 2024.
- P8.** "Synthesis of Chiral Ag, Pd, and Pt Helicoids inside Chiral Silica Mold", Ha, I. H.; Kim, R. M.; Han, J. H.; Im, S. W.; **Jo, J.**; Lee, Y. H.; Lv, J.; Lee, U. C.; Ahn, H.-Y.; Lee, H.-E.; Kim, M.; Nam, K. T., Journal of the American Chemical Society 2024.
- P9.** "Kink-Controlled Gold Nanoparticles for Electrochemical Glucose Oxidation", Choi, S.; Liu, C.; Seo, D. H.; Im, S. W.; Kim, R. M.; **Jo, J.**; Kim, J. W.; Park, G.-S.; Kim, M.; Brinck, T.; Nam, K. T., Nano Letters 2024, 24(15), 4528-4536.
- P10.** "Selective hydrocarbon or oxygenate production in CO₂ electroreduction over metallurgical alloy catalysts", Kim, J.-Y.; Ahn, H. S.; Kim, I.; Hong, D.; Lee, T.; **Jo, J.**; Kim, H.; Kwak, M. K.; Kim, H. G.; Kang, G.; Go, S.; Ryu, W. H.; Lee, G.-D.; Kim, M.; Nam, D.-H.; Park, E. S.; Joo, Y.-C., Nature Synthesis 2024, 3(4), 452-465.
- P11.** "Anomalous optical excitations from arrays of whirlpooled lattice distortions in moire superlattices", Kim, J.; Ko, E.; **Jo, J.**; Kim, M.; Yoo, H.; Son, Y. W.; Cheong, H., Nature Materials 2022, 21(8), 890-895.
- P12.** "Dimensionality reduction and unsupervised clustering for EELS-SI", Ryu, J.; Kim, H.; Kim, R. M.; Kim, S.; **Jo, J.**; Lee, S.; Nam, K. T.; Joo, Y.-C.; Yi, G.-C.; Lee, J.; Kim, M., Ultramicroscopy 2021, 231, 113314.
- P13.** "Strain-Induced Modulation of Localized Surface Plasmon Resonance in Ultrathin Hexagonal Gold Nanoplates", Park, G. S.; Min, K. S.; Kwon, H.; Yoon, S.; Park, S.; Kwon, J. H.; Lee, S.; **Jo, J.**; Kim, M.; Kim, S. K., Advanced Materials 2021, 33(38), 2100653.

FELLOWSHIPS & HONORS

Gordon and Betty Moore Foundation Postdoctoral Fellowship | 2026

Scholarship for Next Generation in Fundamental Disciplines, Seoul National University | 2022-2024

Gwanjung Scholarship, Gwanjung Educational Foundation | 2018-2020

Cheonghap Scholarship, Cheonghap Educational Foundation | 2014-2017

SELECTED PRESENTATIONS

2026 "Monochromated Magnetic-Field-Free Atomic-Resolution STEM: Advancing Quantum Materials Characterization", International Microscopy Congress 21 (IMC21), Liverpool, UK. Oral presentation.

2026 "Direct Nanoscale Mapping of Local Optical Chirality in a Single Chiral Nanoparticle via Photon-induced Near-field Electron Microscopy", International Microscopy Congress 21 (IMC21), Liverpool, UK. Poster presentation.

2026 "New Opportunities with the Advent of Monochromated Magnetic-field-free Atomic-resolution Scanning Transmission Electron Microscopy," Microscopy & Microanalysis (M&M 2026), Milwaukee, WI, USA. Oral presentation.

- 2026** "3D Imaging of Circular Dichroic Plasmonic Near-Fields Using Polarization-Controlled Photo-Induced Near-Field Electron Microscopy (PINEM)," International Microscopy Congress 21 (IMC21), Liverpool, UK. Poster presentation.
- 2025** "3D Imaging of Circular Dichroic Plasmonic Near-Fields Using Polarization-Controlled Photo-Induced Near-Field Electron Microscopy (PINEM)," Asia-Pacific Microscopy Congress (APMC13), Brisbane, Australia. Oral presentation.
- 2024** "Non-Monochromatic Plasmonic Behavior of Metal Nanoparticle/Dielectric Nanorod Composite Material," Materials Research Society Fall Meeting, Boston, MA, USA. Poster presentation.
- 2023** "Unraveling Surface Plasmon Peak Splitting in Silver Nanoparticles Decorated on ZnO Nanorods via Monochromated STEM-EELS Tomography," International Microscopy Conference 20, Busan, Republic of Korea. Poster presentation.
- 2022** "Three-dimensional Tomographic Imaging Simulation of Surface Plasmon Modes of a Chiral Gold Nanoparticle Using STEM-EELS," Materials Research Society Spring Meeting, Honolulu, HI, USA. Oral presentation.
- 2021** "Three-dimensional Tomographic Imaging Simulation of Surface Plasmon Modes on a Chiral Nanoparticle," Korean Society of Microscopy Fall Meeting, Pyeongchang, Republic of Korea. Oral presentation.
- 2021** "Three-dimensional Characterization of Nanoparticles with Simultaneously Acquired Bright-field, Dark-field and Secondary Electron Signals in STEM," PICO 2021, Virtual. Poster presentation.
- 2020** "Understanding the Physics of Interaction Between Surface Plasmons and Dielectric Matters in Ag/ZnO Nanorod System Using STEM-EELS," Materials Research Society Spring/Fall Meeting, Virtual. Oral presentation.
- 2020** "Interpretation of Plasmon Peak Splitting in Silver/ZnO Nanorod Heterostructure via Low-loss EELS Simulation," Korean Ceramic Conference Spring Meeting, Virtual. Oral presentation.
- 2019** "Simulation of Geometric Effect on Plasmons in Silver/ZnO System," European Microscopy Research Society Spring Meeting, Nice, France. Poster presentation.

TEACHING, TRAINING & PROFESSIONAL ACTIVITIES

- Speaker, Thermo Fisher Scientific TEM Workshop, Giheung, Republic of Korea (Mar. 2025).
- Special Lecturer, User Training Workshop, Research Institute of Advanced Materials, Seoul National University - 1.5-hour lecture on monochromated low-loss EELS and practical applications (Jul. 2021; Jan. 2022).
- Participated in Thermo Fisher Scientific Themis Z Application Training, Eindhoven NanoPort Center, Netherlands (Oct. 2022).
- Completed the Gatan EELS & EFTEM Analysis Virtual Training School (Oct. 2020).

AWARDS

- Best Presentation Award for Graduate Students, Korean Society of Microscopy Fall Meeting (2021).
- Best Presentation Award, Materials Research Society Spring/Fall Meeting, Advanced Linear/Nonlinear, Tunable and Quantum Materials for Metasurfaces, Metamaterials and Plasmonics Symposium (2020).
- Commendation for Academic Achievements, President of Alumni Association of the Department (2018).